

Kaushal Rohit

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SUMMARY

AI/ML Developer specializing in vector databases, LLM orchestration, RAG pipelines, and scalable Python backend architecture.

EXPERIENCE

Python Developer Intern

Jan 2026 – Mar 2026

S.M. Soni Infosoft, Vadodara, Gujarat

- Engineered a full-stack Django e-commerce platform (*fabolus-finds*) spanning 9 cascading data models, 20 DB tables, 17 REST endpoints, and 55 HTML templates across 2,500+ lines of Python.
- Architected a multi-app Django monorepo (adminapp, customers, employee, giftkeepers) with custom session-based auth, AJAX-powered dynamic dropdowns, and City → Area → Employee relational hierarchy.
- Built 11 class- and function-based views (7 CBV + 4 FBV) with ORM-backed order lifecycle management covering product catalog, cart checkout, and employee task-assignment workflows.

Technical Instructor — Applied AI, Python & Cloud

Jan 2023 – Feb 2026

IANT – Institute of Advanced Network Technology (Concurrent with Studies), Vadodara, Gujarat

- Trained 100+ students in Applied AI, Python programming, and cloud computing, achieving a 90%+ completion rate for student capstone projects built and tested on AWS EC2.
- Directed hands-on lab curriculum covering AWS infrastructure (EC2, S3, IAM, Lambda, VPC), cloud-based ML deployment workflows, and data pipeline best practices.
- Mentored end-to-end student project development in data analytics and ML, instilling professional standards in code quality, data governance, and systematic debugging.

PROJECTS

Adaptive RAG Pipeline

Python, FAISS, ChromaDB, LangChain, Ollama

- Engineered a dual-backend RAG system indexing **13,917 vectors** with FAISS HNSW (M=32) achieving **0.24ms** search latency and **82.7% Recall@5**, reducing hallucination rate by 20 percentage points vs. baseline.
- Implemented dynamic k-selection governed by real-time sentiment analysis of rolling 5-turn comprehension metrics, boosting context coverage to **60%** and adapting retrieval depth per query complexity.
- Built a hybrid retrieval layer benchmarking FAISS (speed-optimised L2) against ChromaDB HNSW (cosine), with batch embedding generation at 9,217 vectors/sec for educational content search.

GenAI-Powered B2B Insights Analytics Engine

ChromaDB, UMAP, HDBSCAN, sentence-transformers

- Architected a B2B analytics pipeline processing **8,469 interaction records** through sentence-transformers + ChromaDB HNSW indexing at **12.64ms p50** query latency (1,100 vectors/sec insert throughput).
- Applied UMAP dimensionality reduction (384D → 2D) and HDBSCAN clustering achieving **0.5865 silhouette score** with automated 2-cluster discovery and only 0.1% noise — indicating clean cluster boundaries.
- Designed an unsupervised insight discovery pipeline combining 384D sentence embeddings with density-based clustering (HDBSCAN in 0.053s), enabling automatic pattern detection across unstructured B2B text.

Credit Eligibility Prediction System

Python, Scikit-learn, XGBoost, SMOTE

- Built an ensemble ML pipeline (RF + GBM + XGBoost + LR) with SMOTE class rebalancing on a 3,000-record dataset, achieving **0.7932 ROC-AUC** and **0.7391 weighted F1** via 5-fold stratified CV.
- Designed a 26-feature engineering layer including polynomial features (Credit_Score², DTI²) and ratio features (Savings-to-Loan, Income-to-Loan), improving AUC by 4.2% over raw-feature baselines.
- Automated hyperparameter optimisation with RandomizedSearchCV across 4 model families; per-class results: Approved F1=0.76, Rejected F1=0.72 on a balanced 52%/48% post-SMOTE distribution.

AI-Powered Mock Interview Platform

Next.js 16, TypeScript, Groq API, Deepgram STT

- Architected a real-time AI interview platform in **3,753 LoC** (26 source files) with 6 RESTful API routes covering PDF resume parsing, LLM-powered question generation, and OTP authentication.
- Designed a **303-line finite state machine** managing adaptive 8-question sessions with 90-second response windows, 4 question types (Technical, Conceptual, Behavioral, Scenario), and 5-dimension scoring.
- Integrated Groq LLM for sub-second adaptive difficulty scaling and Deepgram STT for real-time voice transcription, with performance dashboards built using Recharts data visualisation.

SKILLS

- AI/ML & RAG Engineering:** LLMs, Prompt Engineering, FAISS, ChromaDB, LangChain, Semantic Search, SMOTE, UMAP, HDBSCAN, Scikit-learn, XGBoost.
- Backend & Data Systems:** Python (Primary), TypeScript, Django, Next.js, REST APIs, SQL, Pandas, NumPy.
- Cloud & Infrastructure:** AWS (EC2, S3, IAM, Lambda, VPC), Ollama, Groq API, Deepgram.
- Core Competencies:** Feature Engineering, Data Processing, Context Assembly, Hallucination Mitigation.

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science Engineering

April 2026

Drs. Kiran & Pallavi Patel Global University (KSETAR), Vadodara, Gujarat

Diploma in Computer Science Engineering

2023

Government Polytechnic, Ahmedabad, Gujarat